

**DecodeLabs – Data Analytics
Internship
Project 2: Exploratory Data
Analysis**

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DecodeLabs Internship – Project 2

Exploratory Data Analysis (EDA)

1. Project Overview

This project focuses on performing Exploratory Data Analysis (EDA) on the cleaned e-commerce dataset. The objective was to understand the characteristics of the data, identify trends and patterns, analyze products and order-related information, detect potential outliers, and extract meaningful observations.

2. Project Objective

The main objectives were:

- Understand the dataset using descriptive statistics.
- Analyze important trends.
- Examine product performance.
- Analyze order status.
- Analyze payment methods.
- Identify potential outliers.
- Create meaningful visualizations.
- Extract key observations from the data.

3. Dataset Description

The cleaned dataset contains e-commerce/order information including:

- Order ID
- Date
- Product
- Quantity
- Unit Price
- Total Price
- Payment Method
- Order Status
- Coupon Code
- Other order related attributes

4. Exploratory Data Analysis

4.1 Basic Statistics

Descriptive statistics were calculated to understand the numerical characteristics of the dataset.

The analysis included:

- Count
- Mean
- Median
- Minimum
- Maximum
- Supporting statistical measures

These statistics provide an overall understanding of the dataset and help identify unusual values.

4.2 Trend Analysis

The data was analyzed over time to identify changes in:

- Number of orders
- Sales
- Monthly performance
- Yearly performance

Trend analysis helps identify periods of higher or lower business activity.

4.3 Product Analysis

Products were analyzed based on:

- Number of orders
- Quantity sold
- Total sales
- Average order value

This helps identify high-performing and lower-performing products.

4.4 Order Status Analysis

Order status was analyzed to understand the distribution of different order outcomes.

This provides an overview of the operational status of orders.

4.5 Payment Method Analysis

Different payment methods were compared based on order activity and sales.

This helps understand customer payment preferences and their contribution to sales.

5. Outlier Analysis

Potential outliers were identified using the Interquartile Range (IQR) method.

IQR Formula

$$\text{IQR} = Q3 - Q1$$

Lower Bound:

$$Q1 - 1.5 \times \text{IQR}$$

Upper Bound:

$$Q3 + 1.5 \times IQR$$

Values outside these boundaries were identified as potential outliers.

Outliers were not automatically removed because they may represent genuine high-value or unusual transactions.

6. Visualizations

Charts were created to support the analysis.

The visualizations include:

- Monthly order trends
- Monthly sales trends
- Product performance
- Order status distribution
- Other relevant comparisons

Visualizations make patterns easier to understand than raw numerical tables.

7. Key Observations

The EDA helped identify:

- High-performing products
- Important sales periods
- Order patterns
- Payment method activity
- Unusual observations
- Overall sales behavior

Detailed observations are available in the Excel workbook.

8. Tools Used

- Microsoft Excel
- Exploratory Data Analysis
- Descriptive Statistics
- Data Visualization
- Outlier Analysis

9. Project Output

The final workbook contains:

1. Clean Data

2. Basic Statistics
3. Trend Analysis
4. Outlier Analysis
5. Key Observations
6. Charts

10. Skills Demonstrated

- Exploratory Data Analysis
- Descriptive Statistics
- Trend Analysis
- Outlier Detection
- Data Visualization
- Microsoft Excel
- Business Insight Generation

11. Conclusion

This project provided practical experience in exploring a dataset and identifying useful patterns before performing advanced analysis.

EDA helped transform raw numerical data into understandable trends, comparisons, and observations that can support business decision-making.